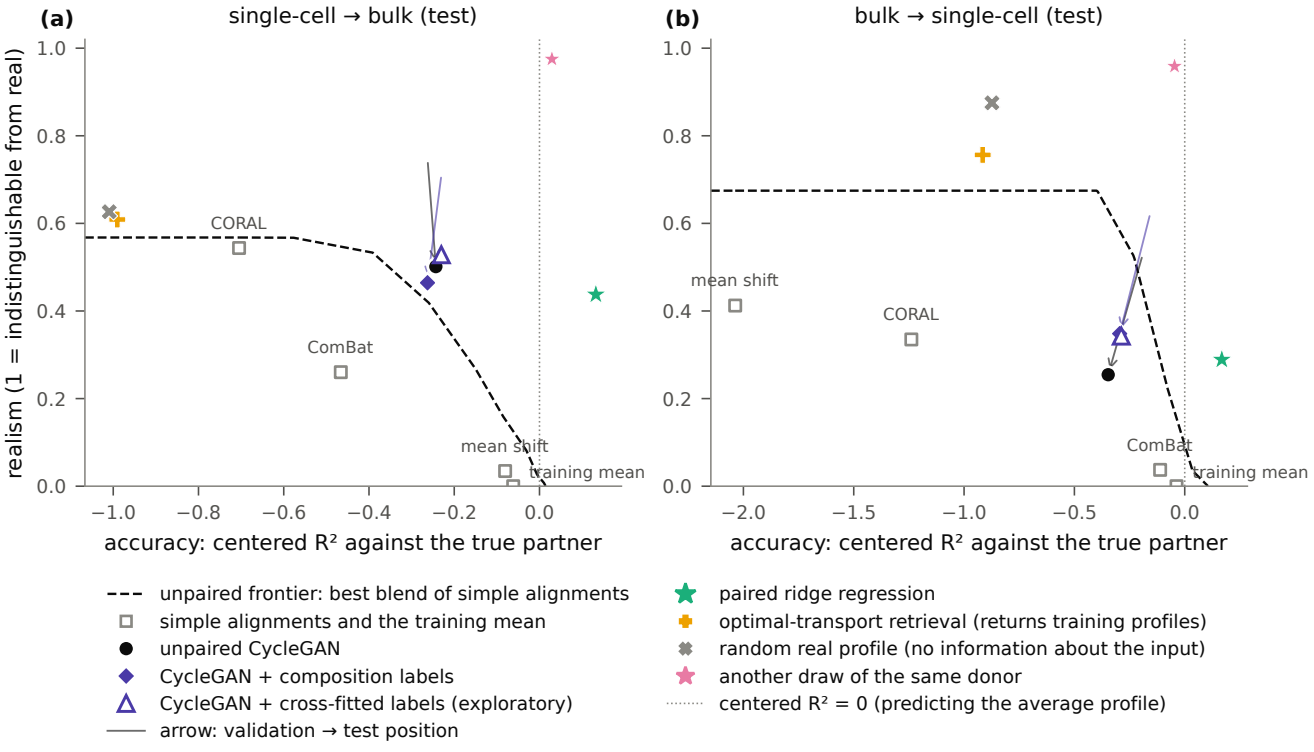
(a)
single-cell → bulk (test)
realism (1 = indistinguishable from real)
accuracy: centered R² against the true partner
CORAL
ComBat
mean shift
training mean
(b)
bulk → single-cell (test)
accuracy: centered R² against the true partner
mean shift
CORAL
ComBat
training mean
unpaired frontier: best blend of simple alignments
simple alignments and the training mean
unpaired CycleGAN
CycleGAN + composition labels
CycleGAN + cross-fitted labels (exploratory)
arrow: validation → test position
paired ridge regression
optimal-transport retrieval (returns training profiles)
random real profile (no information about the input)
another draw of the same donor
centered R² = 0 (predicting the average profile)